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UK Visa Status: Student Visa. Eligible for Graduate Visa (2 Years Full Working Rights)

STATEMENT

Applied AI Engineer with 3 years of industry experience building and running production GenAI systems for internal customers at HDFC Asset Management, taking each one from scoping with the business team through release, security review and iteration on real usage: **RAG** over PostgreSQL with **pgvector**, tool-calling **agents**, and a self-hosted **Llama-3** model on a GPU instance alongside **Azure OpenAI**, allocated by cost. Started as the only engineer besides my manager, owned the systems end to end, and later acted as de facto team lead for up to 4 interns; twice recognised internally with Best Team Project. Recently completed an MSc in Data Science at the University of Bristol, with a dissertation built with feedback from AstraZeneca scientists. Seeking to bring this ownership-first approach to Callosum's Applied AI team.

EDUCATION

University of Bristol – MSc. Data Science (awaiting result) September 2025 – September 2026
NMIMS B.Tech Information Technology (Dual Degree) June 2017 – April 2022
NMIMS MBA Technology Management (Dual Degree) June 2017 – April 2022

WORK EXPERIENCE

HDFC Asset Management Company Ltd. July 2022 – July 2025

AI Solutions Engineer / Forward Deployed Engineer

Customer-Embedded Retrieval, Agents & Model Serving

- Embedded with Compliance, Client Service and Investment teams to scope, design and build, end to end, the GenAI applications they used (frontend, backend, **RAG** logic, vector database and embeddings pipeline; the first systems were designed with just my manager): retrieval pipelines using **LlamaIndex** with **Azure OpenAI** embeddings and a separate **pgvector** database for each department so access stayed siloed and similar content from different departments was not confused, delivered as a **ReactJS** frontend and Python **FastAPI** microservices.
- Built **agentic tool-calling** workflows where AI agents call **Python** functions as tools to run **SQL**-backed fund-comparison calculations and generate **PDF** comparison reports on user request.
- Built a **Python** email-processing pipeline that parses over **300 daily emails** from credit rating agencies into structured data, fuzzy-matches them to the companies analysts track and sends a daily summary, **saving 60 manual analyst hours a month**; replaced **AWS Textract** with **Azure OpenAI** when a rating agency began sending emails as images, which also improved accuracy; scheduled on **AWS SageMaker** via **EventBridge** and **Lambda**.
- Allocated models by cost and workload: open-source **Llama-3 self-hosted on a GPU EC2 instance** for the largest department (Client Service) and stronger **Azure OpenAI** outputs for decision makers, with the ability to switch models. Released to real users and iterated on what broke; guardrailed responses through **system prompts**, versioned every prompt change in **Git** and had the Compliance department review each application before release.

Stakeholder Discovery, Data & Integration

- Worked directly with stakeholders in Client Service, Compliance, Investment and marketing to turn requirements into working tools: developed **Power BI** dashboards for fortnightly credit report analysis, migrating financial analysts from **Excel to Power BI** and improving analyst efficiency by **10 hours a week**; then recreated the dashboards in the **ReactJS** frontend so the wider organisation could use the data without Power BI licensing costs.

- Designed a **Normalized SQL Schema** (**PostgreSQL**) and wrote **Python** ingestion scripts for the **Morningstar API** covering every fund in the Indian mutual fund market (2023–2025), refreshed fortnightly (fixed income) and monthly (equity); **integrated it with the legacy data** the company already held in analysts' Excel files; developed **Optimized SQL** queries, **SQL Functions** and **Procedures**.

Enterprise Deployment, Security & Leadership

- Designed the infrastructure architecture for internal analytics platforms and GenAI applications on **AWS** utilizing core services such as **EC2, S3, ECR and Lambda**, with **Docker** microservices (FastAPI, uvicorn), separate frontend and backend repositories and containers, and a one-click **Jenkins** and **GitLab** CI/CD pipeline for dev and prod.
- Implemented **RBAC** for the GenAI apps with a Python authentication function against the company **LDAP** server; restricted access to company private IPs using **AWS Security Groups**; deployed an **Nginx** reverse proxy, one of the fixes that took the applications from repeatedly failing to passing the **VAPT** (vulnerability assessment and penetration testing); coordinated with the security team on port whitelisting and development safeguards.
- Acted as **de facto team lead** for up to 4 interns in my final year, alongside a senior engineer and a dedicated frontend developer; recognised with **Best Team Project** twice (2024 rating-email automation, 2025 department RAG chatbot).

PROJECTS

- **University of Bristol MSc. Data Science Dissertation in partnership with AstraZeneca:** CellLineSelector (Team project, co-built; I designed the RAG-based AI assistant and worked on the scoring algorithm and data preprocessing for a tool, built as FastAPI microservices with a React and TypeScript frontend, that ranks cancer cell lines by target inclusion and exclusion genes; interacted directly with AstraZeneca scientists, whose feedback shaped and validated the ranking.) GitHub: https://github.com/Arishtanemi4/az-team25.git | Demo: http://celllineselector-frontend-408937187029.s3-website.eu-west-2.amazonaws.com/
- **Wingman:** AI dating-coach web app with 108 LLM personas across 12 archetypes (ReactJS, Python FastAPI, Kimi K2 and DeepSeek V4 Pro via NVIDIA NIM, selectable per request). Behaviour steered by prompts; LLM output treated as untrusted: validated with Pydantic, repaired and cached by input hash; a 40 requests-per-minute rate limit shaped the design (12 calls instead of 108, serialised calls, streamed results). GitHub: https://github.com/Arishtanemi4/wingman.githttps://github.com/Arishtanemi4/wingman.git
- **University of Bristol MSc Applied Data Science Group Project:** VisionCaptioner (Team project. Architecture study comparing CNN/ViT/ResNet encoders and LSTM/attention decoders for generating image descriptions, benchmarked against BLIP/SigLIP and commercial VLMs.) GitHub: https://github.com/Arishtanemi4/ata-team30.git

ACHIEVEMENTS

- **Project Hack 27**: Rolls Royce Derby 4th Place winner. (Human Centric Data Analytics tool to predict the competition winner based on human wellbeing and psychological temperament survey data.) | GitHub: https://github.com/Arishtanemi4/team-4b.git
- **Encode Vibe Coding Hackathon:** Codeplain.ai track 2nd place winner. (Wingman.ai – An AI powered dating app simulation using codeplain's spec-driven framework) | GitHub: https://github.com/Arishtanemi4/wingman-codeplain.git
- **University of Bristol Datathon sponsored by Lloyd's Bank:** 3rd Place winner. (Kaggle Credit card fraud detection problem.)

SKILLS

- **Tools:** Jupyter, Anaconda, AWS (EC2 incl. GPU instances, S3, ECR, Lambda, SageMaker), Azure OpenAI, LlamaIndex, Claude Code, Docker, Jenkins, GitLab, Git, Nginx, Linux, Power BI, VS Code.
- **Languages:** Python (pandas, scikit-learn, FastAPI, Pydantic), SQL, TypeScript (React frontend, CellLineSelector), R (dplyr, tidyverse, ggplot2), NodeJS (express, ReactJS), Shell
- **Databases:** PostgreSQL (pgvector, SQL functions and procedures), SQLite